MATH 152
Mrs. Bonny Tighe

QUIZ 9A

25 points
11.9

NAME Solutions

TA_____Wed Wed 11/14/2018

4pts

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{9x^2}{1+3x^3}$ Find the interval of convergence for this series only .

$$\frac{9x^2}{1-(-3x^3)} = \sum_{n=0}^{\infty} 3^2x^2(-3x^3)^n = \boxed{\sum_{n=0}^{\infty}(-1)^n 3^{n+2} x^{3n+2}}$$

$$|r| = |-3x^3| < 1 \qquad |x^3| < 1/3 \qquad |x| < \sqrt[3]{1/3}$$

$$I = \left(-\sqrt[3]{1/3}, \sqrt[3]{1/3}\right)$$

4pts

b) $f(x) = \ln(x^3+4)$

$$f'(x) = \frac{3x^2}{4+x^3} = \frac{\frac{3}{4}x^2}{1-\left(-\frac{x^3}{4}\right)} =$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{3}{4}x^2\left(-\frac{x^3}{4}\right)^n = \sum_{n=0}^{\infty} 3\frac{(-1)^n x^{3n+2}}{4^{n+1}}$$

$$\int f'(x)dx = \boxed{C + \sum_{n=0}^{\infty} \frac{3(-1)^n x^{3n+3}}{4^{n+1}(3n+3)}}$$

4pts

c) $f(x) = \dfrac{x}{(9-x^2)^2}$

$$\int f(x)\,dx = \int (9-x^2)^{-2}\, x\,dx \qquad u = 9-x^2 \quad du = -2x\,dx$$

$$-\frac{1}{2}\int u^{-2}\,du = -\tfrac{1}{2}\cdot\frac{1}{-1}u^{-1} = \frac{1}{2(9-x^2)} = \frac{1/2}{9-x^2} = \frac{\frac{1}{2}\cdot\frac{1}{9}}{1-\frac{x^2}{9}}$$

$$\int f(x)\,dx = \sum_{n=0}^{\infty} \frac{1}{2}\cdot\frac{1}{9}\left(\frac{x^2}{9}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2}\,\frac{x^{2n}}{9^{n+1}}$$

$$\frac{d}{dx}\int f(x)\,dx = \sum_{n=1}^{\infty} \frac{1}{\cancel{2}}\cdot\frac{\cancel{2}n\,x^{2n-1}}{9^{n+1}}$$

$$\boxed{\sum_{n=0}^{\infty} \frac{(n+1)\,x^{2n+1}}{9^{n+2}}}$$

2. Approximate the integral using power series and 4 non-zero terms.

4pts

$$\int_0^{0.2} \frac{4}{1-2x^2}\,dx \qquad \int_0^{0.2} \sum_{n=0}^{\infty} 4(2x^2)^n\,dx = \int_0^{0.2} \sum_{n=0}^{\infty} \left(2^{n+2}x^{2n}\right)dx$$

$$\sum_{n=0}^{\infty} \frac{2^{n+2}x^{2n+1}}{2n+1}\Bigg]_0^{0.2} = \frac{2^2x}{1} + \frac{2^3x^3}{3} + \frac{2^4x^5}{5} + \frac{2^5x^7}{7}\Bigg]_0^{0.2}$$

$$\left(4(0.2) + \frac{8(0.2)^3}{3} + \frac{16(0.2)^5}{5} + \frac{32(0.2)^7}{7}\right) - (0)$$

4pts

3. Find $f'(x)$ for $f(x)=\sum_{k=0}^{\infty}\frac{(-1)^k 4^k \; x^{2k}}{(1\cdot 5\cdot 9\cdots(4k-3))}$. Then find the radius and interval of convergence.

$$f'(x)=\sum_{k=1}^{\infty}\frac{(-1)^k 4^k \; 2k\, x^{2k-1}}{(1\cdot 5\cdot 9\cdots(4k-3))}$$

$$\lim_{k\to\infty}\left|\frac{4\cdot 4^k\, 2(k+1)\, x^2\, \cancel{x^{2k-1}}}{(1\cdot 5\cdot 9\cdots(4k-3)(4k+1))}\cdot\frac{(1\cdot 5\cdot 9\cdots(4k-3))}{\cancel{4^k}\, 2k\, \cancel{x^{2k-1}}}\right|$$

$$\lim_{k\to\infty}\left|\frac{8(k+1)}{(4k+1)(2k)}\, x^2\right| = 0 < 1 \text{ always} \quad R=\infty$$

$$I=(-\infty,\infty)$$

5pts

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x)=\frac{1-5x}{1+2x-3x^2}=\frac{1-5x}{(1+3x)(1-x)}$

$$\frac{A}{1+3x}+\frac{B}{1-x}=\frac{1-5x}{(1+3x)(1-x)}$$

$$A(1-x)+B(1+3x)=1-5x$$

$x=1$ $\quad 4B=-4 \quad B=-1$

$x=-1/3$ $\quad 4/3\,A = 1-5(-1/3)$

$4/3\,A = 1+\frac{5}{3}=\frac{8}{3}\cdot\frac{3}{4}$

$A=2$

$$\frac{2}{1-(-3x)}+\frac{-1}{1-x}$$

$$\sum_{n=0}^{\infty}2(-3x)^n+\sum_{n=0}^{\infty}-1(x)^n$$

$$\sum_{n=0}^{\infty}2(-1)^n3^nx^n-\sum_{n=0}^{\infty}x^n$$

or $\sum_{n=0}^{\infty}x^n\left[2(-1)^n3^n-1\right]$

$|r|=|-3x|<1 \quad |x|<1/3$

$I=-1/3,\ 1/3$

$|r|=|x|<1 \quad I=-1,1$

$$I=(-1/3,\ 1/3)$$

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QUIZ 9

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NAME Solution

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4pts

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{x}{5 - x^3}$ Find the interval of convergence for this series only .

$$f(x) = \frac{\frac{x}{5}}{1 - \frac{x^3}{5}} = \sum_{n=0}^{\infty} \frac{x}{5}\left(\frac{x^3}{5}\right)^n = \boxed{\sum_{n=0}^{\infty} \frac{x^{3n+1}}{5^{n+1}}}$$

$$r = \left|\frac{x^3}{5}\right| < 1 \qquad |x^3| < 5 \qquad |x| < \sqrt[3]{5}$$

$$I = (-\sqrt[3]{5}, \sqrt[3]{5})$$

4pts

b) $f(x) = \ln(3x+1)$

$$f'(x) = \frac{3}{1+3x} = \frac{3}{1-(-3x)} = \sum_{n=0}^{\infty} 3(-3x)^n$$

$$\boxed{f'(x) = \sum_{n=0}^{\infty} (-1)^n 3^{n+1} x^n}$$

$$\int f'(x)\,dx = \boxed{C + \sum_{n=0}^{\infty} (-1)^n \frac{3^{n+1} x^{n+1}}{n+1}}$$

4pts

c) $f(x) = \dfrac{x}{(1+x^2)^2}$

$\int (1+x^2)^{-2} x\,dx \qquad u = 1+x^2 \qquad du = 2x\,dx$

$\frac{1}{2}\int u^{-2}\,du = -\frac{1}{2}\cdot\frac{1}{1}u^{-1} = \dfrac{-1}{2(1+x^2)}$

$\int f(x)\,dx = \dfrac{-1/2}{1-(-x^2)} = \sum_{n=0}^{\infty} -1/2\,(-x^2)^n = \sum_{n=0}^{\infty} \dfrac{(-1)^{n+1}x^{2n}}{2}$

$\dfrac{d}{dx}\int f(x)\,dx = \sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}\cancel{2}n\,x^{2n-1}}{\cancel{2}} = \sum_{n=1}^{\infty} (-1)^{n+1} n\,(x)^{2n-1}$

$$\boxed{\sum_{n=0}^{\infty} (-1)^n (n+1)\,x^{2n+1}}$$

2. Approximate the integral using power series and 4 non-zero terms.

4pts

$\int_0^{0.2} \dfrac{4x}{1-2x^2}\,dx = \int_0^{0.2} \sum_{n=0}^{\infty} 4x(2x^2)^n\,dx = \int_0^{0.2} \sum_{n=0}^{\infty} 2^{n+2} x^{2n+1}\,dx$

$\sum_{n=0}^{\infty} \dfrac{2^{n+2}x^{2n+2}}{2n+2}\Big]_0^{0.2} = \dfrac{2^2x^2}{2} + \dfrac{2^3x^4}{4} + \dfrac{2^4x^6}{6} + \dfrac{2^5x^8}{8}\Big]_0^{0.2}$

$\left(\dfrac{4(0.2)^2}{2} + \dfrac{8(0.2)^4}{4} + \dfrac{16(0.2)^6}{6} + \dfrac{32(0.2)^8}{8}\right) - (0)$

4pts

3. Find $f'(x)$ for $f(x)=\sum_{k=0}^{\infty}\frac{(-1)^k 3^k \; x^{k+2}}{(1\cdot 5\cdot 9\cdots(4k-3))}$. Then find the radius and interval of convergence.

$$f'(x)=\sum_{k=0}^{\infty}\frac{(-1)^k 3^k (k+2)\, x^{k+1}}{(1\cdot 5\cdot 9\cdots(4k-3))}$$

$$\lim_{k\to\infty}\left|\frac{3\cdot 3^k (k+3)\, x\cdot x^{k+1}}{(1\cdot 5\cdot 9\cdots(4k-3)(4k+1))}\cdot\frac{(1\cdot 5\cdot 9\cdots(4k-3))}{3^k (k+2)\, x^{k+1}}\right|$$

$$\lim_{k\to\infty}\left|\frac{3(k+3)}{(4k+1)(k+2)}\cdot x\right| = 0 < 1 \text{ always}$$

$R=\infty$

$I=(-\infty,\infty)$

5pts

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x)=\frac{2+x}{1-4x+3x^2}$

$$\frac{2+x}{(1-3x)(1-x)}$$

$$\frac{A}{1-3x}+\frac{B}{1-x}=\frac{2+x}{(1-3x)(1-x)}$$

$A(1-x)+B(1-3x)=2+x$

$x=1 \qquad B(-2)=3 \qquad B=-3/2$

$x=1/3 \qquad A(-2/3)=8/3$

$A=8/3(-3/2)=-4$

$$\frac{-4}{1-3x}+\frac{-3/2}{1-x}$$

$$\sum_{n=0}^{\infty}-4(+3x)^n+\sum_{n=0}^{\infty}-3/2\, x^n$$

$$\sum_{n=0}^{\infty}\left(-4(3^n x^n)-3/2\, x^n\right)=\sum_{n=0}^{\infty}x^n\left[-4\cdot 3^n-3/2\right]$$

$|r|=|3x|<1$

$|x|<1/3$

$I=(-1/3,1/3)$

$|r|=|x|<1 \qquad I=(-1,1)$

so $I=(-1/3,1/3)$